LYON CAPITAL

BESS – Battery Energy Storage System

Preliminary Assessment of the Brazilian Market Potential

January 2026

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Section 1

Lyon Capital

LYON CAPITAL

Lyon Capital

- Lyon Capital is an alternative investment management company, recognized for its track record in highly profitable transactions in infrastructure and real estate.
- Lyon team is fully engaged in every stage of the investment processes and set itself apart for its capacity to deal with complex financial situations, create innovative and effective solutions and originate unique investment ideas.

Main Areas

Real Estate

- Track record with more than 200 projects developed in the industrial, logistics, shopping centers, retail and commercial buildings sectors.
- Partners led negotiations for the main real estate projects in the country, including Santander Tower, JK Iguatemi Mall (SP) and Petrobras headquarter (RJ).
- Brought to Brazil innovations such as Built-to-Suit.

Infrastructure

- Partners have overseen a CAPEX of over R$ 8 billion in projects, including highways, shipyards, transmission lines, among others.
- Recognized presence in the port and energy sectors.
- Focused on investments in Power Generation, Transmission Lines, Highways, Ports and Logistics Terminals.

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Lyon Capital – Selected projects

CAPEX	USD[1] 881 mm
Total Area	500,000 m²
Conclusion	2012

Complexo JK

Complexo JK has 500,000 m² area and is one of the main landmark of São Paulo city. The complex is integrated by five blocks, which includes Santander Tower, two business towers, Johnson & Johnson building, Santander theater and JK Iguatemi mall.

CAPEX	USD[1] 470 mm
Total Area	550,000 m²

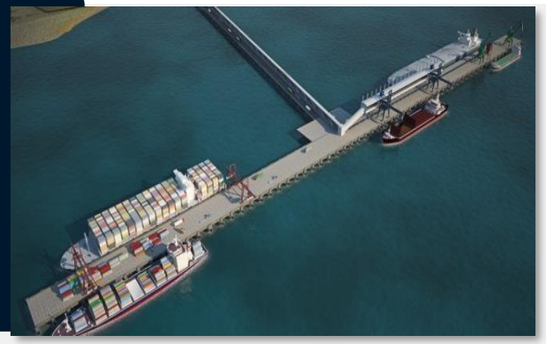
Estaleiro Rio Grande

Rio Grande Shipyard has the largest dry dock in Latin America, with the capacity to build or maintain two FPSOs simultaneously.

CAPEX	USD[1] 245 mm
Soccer Games	43,000 seats
Spectators	55,000
Conclusion	2014

Allianz Parque

One of the most important arenas in Brazil, standing out for its modern and multipurpose characteristics. The arena has been entirely adapted to host sports tournaments, concerts and corporate events. Today, the arena hosts an average of 30 international-level concerts annually.



CAPEX	USD[1] 600 mm
Capacity	24mm ton/year

Porto São Luís

Multipurpose port with the capacity of receiving 400 vessels per year, including capesize. Designed to move grains, fertilizers, minerals and general cargo. The project was sold to COSAN and is currently being restructured to support it full capacity exclusively iron ores operations.

[1] USD at conclusion date

Lyon Capital – Infrastructure Projects Timeline

More than 35 years of experience in the market and with a history of successful projects and above-market returns.

2009

BR Vias S.A. Highway

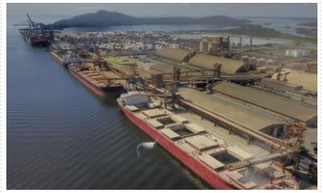
2013

Estaleiro Rio Grande Shipyard

2014

Allianz Parque Arena

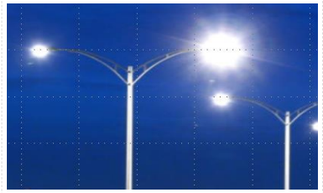
2020



São Luís Port

2021

Transmission Lines



Public Lightning

Solar Energy Generation HUB

2022

Biomass Thermal Plant

2028

Liquified Natural Gas Terminal

Industrial and Storage HUB

2029

Maranhão Port Complex

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Section 2

BESS – Battery Energy Storage System

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BESS: The Invisible Infrastructure Enabling Energy Transition

Brazil's power sector has solved renewable generation. The current challenge isn't cheap energy—it's available power at the right time.

LRCAP 2026 (Government auction) marks the formal entry of storage as a regulated capacity asset, transforming BESS from complementary solution to critical infrastructure.

Renewable Expansion

Wind and solar reached 34% of electricity generation in 2025, projected to exceed 40% by 2030, intensifying grid management challenges.

Operational Constraints

In 2025, curtailment reached 20 TWh (double 2024 levels), with 20% of solar output cut in August. Grid congestion and transmission bottlenecks cost generators BRL 1.7 billion.

Storage Solution

BESS delivers power on command, not just intermittent energy.

Sources: Rystad Energy (2025), PV Magazine Brazil (2025), Ember Energy (2025)

SYSTEM CONTEXT

The Paradigm Shift: From Energy to Flexibility

Traditional power systems were designed around predictable, dispatchable generation. Renewable expansion has fundamentally changed this equation.

The Core Problem:

Energy Abundance ≠ System Reliability

Brazil has abundant renewable energy, but lacks the operational flexibility to deliver it when and where needed.

What Changed:

- Generation became intermittent and weather-dependent
- Load patterns didn't change to match generation
- Grid infrastructure became the bottleneck
- System operators lost direct control over supply timing

What's Needed Now:

From "How much energy?" to "When is power available?"

The critical question shifted from total MWh to dispatchable MW at critical hours.

From Generation Capacity to Operational Flexibility

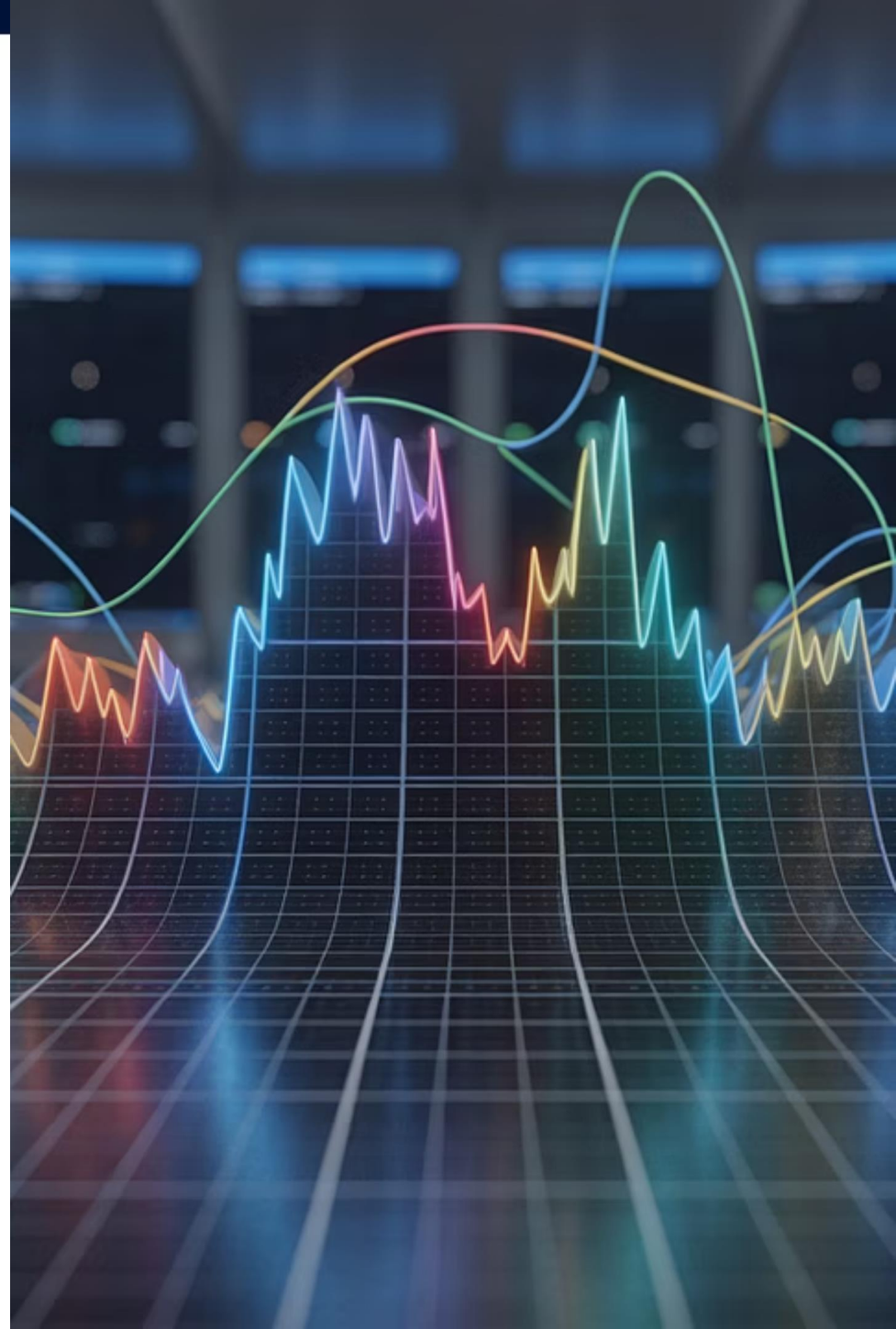
The system needs assets that respond to commands, not weather.

From Energy Markets to Capacity Markets

Value migrates from production volume to availability and reliability.

Key Insight:

> "The system doesn't need more energy—it needs controllable power when demand peaks and renewables don't."



TECHNICAL OVERVIEW

BESS: More Than Batteries—An Electrical Control Asset

A Battery Energy Storage System (BESS) is fundamentally different from any other power system asset.

Main Components:

- Batteries (predominantly Li-ion LFP)
- PCS - Power Conversion System (bidirectional inverters)
- BMS - Battery Management System
- EMS - Energy Management System
- Auxiliary systems: HVAC, protection, fire suppression, SCADA

Technical Differentiators:

Four-Quadrant Operation

Can supply or absorb active AND reactive power simultaneously

Millisecond Response

Unmatched speed for frequency and voltage control

Climate Independence

Doesn't depend on sun, wind, or fuel—delivers power on command

Operational Versatility

Acts as load, generation, and reserve as system needs

> "No thermal, hydro, or renewable delivers this combination of attributes simultaneously."

Section 3

BESS Market

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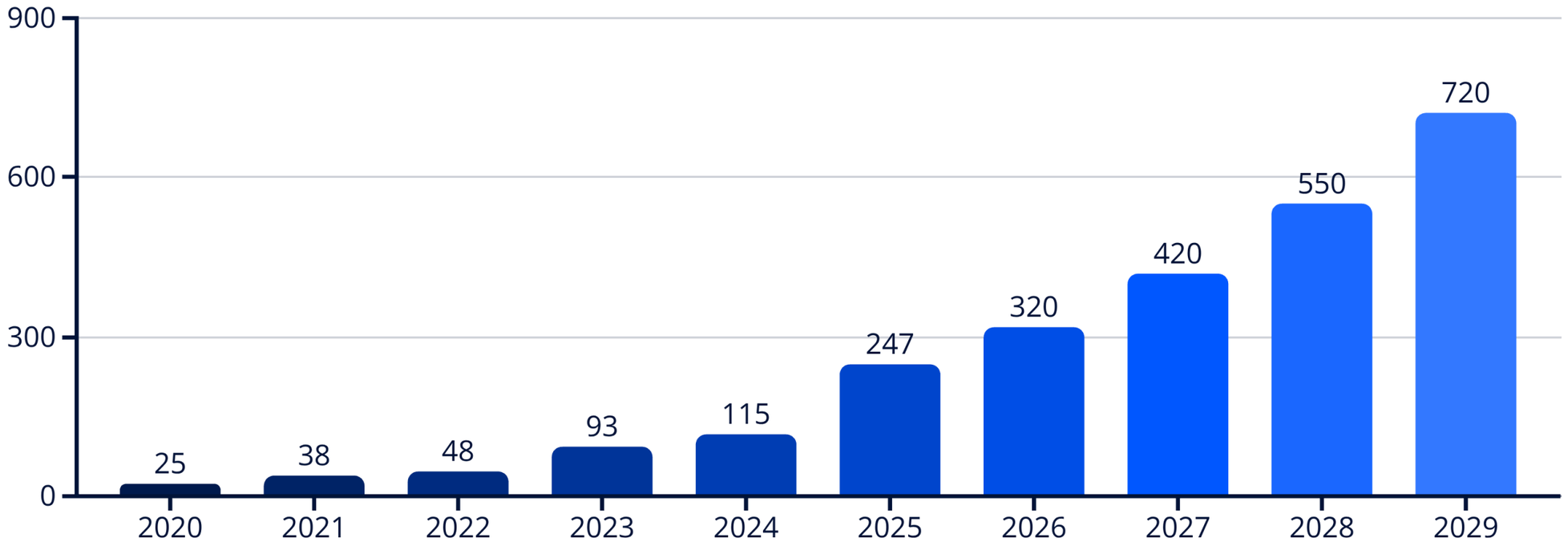
GLOBAL MARKET

Annual Global Energy Storage Deployments (GWh)

The global energy storage market is experiencing explosive growth, with annual deployments doubling to over 240 GWh in 2025. This rapid expansion signifies a fundamental shift, moving energy storage from pilot projects to mainstream infrastructure investment worldwide.

CAGR of 42% (2020-2029) - Growth accelerating in capacity, project scale, and geographic diversity.



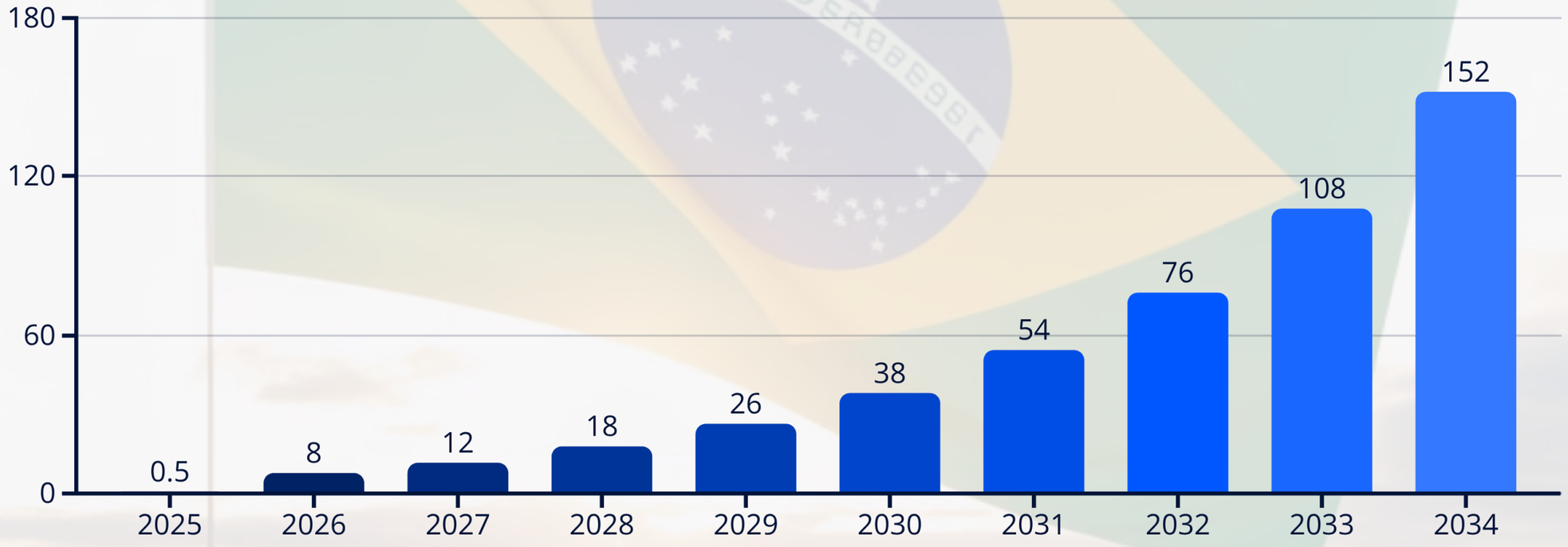
Source: BloombergNEF 2H 2025 Energy Storage Market Outlook. Projections based on continued market growth driven by utility-scale projects and renewable integration requirements.

BRAZIL

Brazilian Energy Storage Market

Brazil is positioned for explosive growth in energy storage, driven by its first dedicated auction (LRCAP 2026) and rapid renewable expansion. The market was valued at USD 217 million in 2025 and is projected to reach USD 4.5 billion by 2034, with the LRCAP 2026 auction targeting 2 GW (~8 GWh) of initial capacity.

CAGR of 40% (2025-2034) - Growth accelerated by regulatory auctions, renewable integration, and operational grid requirements.



Sources: U.S. International Trade Administration, IMARC Group Brazil Energy Storage Market Report 2026-2034. Capacity projections based on LRCAP auction targets and market growth trends.

BRAZIL

Why the Timing Is Now in Brazil

Brazil is experiencing the convergence of three critical factors that create ideal conditions for utility-scale storage deployment: regulatory momentum, operational necessity, and technology maturity. This alignment is rare and time-sensitive.

Regulatory Breakthrough

The Ministry of Mines and Energy (MME) has launched public consultation for Brazil's **first dedicated battery storage auction (LRCAP 2026 – Armazenamento)**, establishing formal procurement mechanisms and revenue certainty for developers.

Simultaneously, ANEEL is advancing specific regulations through **Public Consultation 39/2023**, addressing storage and reversible plant integration into the regulatory framework.

Operational Pressure

The National Interconnected System (SIN) faces mounting operational challenges as renewable penetration increases. ONS is actively studying curtailment scenarios and generation restrictions projected through 2026-2029.

Grid congestion in the Northeast, transmission bottlenecks, and renewable firming requirements are creating tangible economic pain that storage can directly address.

Technology Readiness

Battery technologies—both lithium-ion and flow batteries—have reached commercial maturity with proven track records at gigawatt-hour scale internationally.

Cost trajectories have declined substantially, project financing frameworks exist, and Brazilian engineering capability has developed through adjacent renewable projects.

> "The confluence of regulatory clarity, system operator need, and proven technology creates a narrow window for first-movers to establish position in Brazil's emerging storage market."

What the Power System "Buys" When It Contracts Storage

Energy storage isn't a single product—it's a portfolio of critical grid services bundled into dispatchable infrastructure. When system operators and utilities procure storage capacity, they're purchasing operational flexibility that traditional generation cannot provide at comparable speed or precision.

Firm Capacity (MW)

Reliable power delivery during peak demand periods and system adequacy requirements

Operational Flexibility

Rapid ramping capability (up/down) and grid support services for managing variability

Ancillary Services

Frequency regulation, voltage support, congestion management, and transmission/distribution reinforcement

These services work together to create a more resilient, efficient power system. Storage acts as a shock absorber—smoothing out fluctuations, preventing congestion, and ensuring that renewable energy can be fully integrated without compromising reliability. In markets with high renewable penetration, these capabilities are becoming indispensable.

> "Battery energy storage systems provide the operational agility that modern grids require to balance intermittent renewables while maintaining system stability and reliability."

Section 4

Battery Storage Technology Alternatives

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Primary Storage Solutions: An Infrastructure Perspective

Different storage technologies excel at different duration ranges and operational profiles. Understanding this spectrum is critical for matching technology to application and ensuring optimal system design.

It looks viable for Brazil

Lithium-Ion (LFP/NMC)

Dominant in **1-4 hour** applications due to cost, supply chain maturity, and high efficiency. Best for frequency regulation and daily cycling.

Flow Batteries (VFB)

Strong in **4-10+ hour** duration with superior cycling capability and intrinsic safety. Ideal for extended discharge and heavy-duty operations.

It doesn't look viable for Brazil

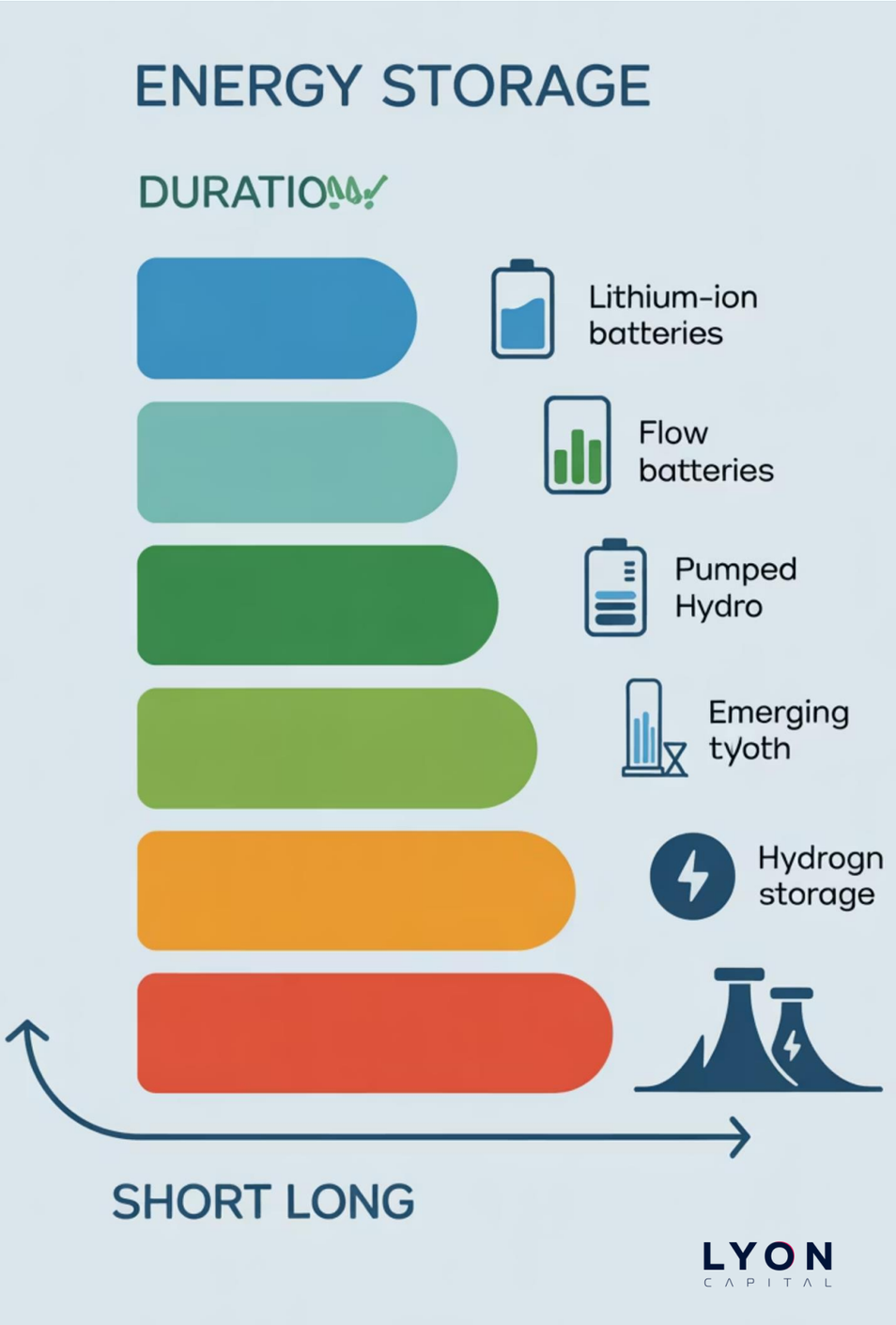
Pumped Hydro

Largest deployed storage technology globally at **8-24+ hours**, where geography permits. Long-duration seasonal storage at massive scale.

Emerging Technologies

CAES, thermal storage, hydrogen for **days-to-weeks** duration, typically with lower round-trip efficiency but enabling multi-day flexibility.

This duration spectrum creates distinct market segments. Short-duration lithium-ion dominates where rapid response and daily cycling are paramount. Long-duration technologies like flow batteries and pumped hydro become economically compelling when discharge duration extends beyond four hours and systems require thousands of deep cycles over project lifetimes.



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Lithium-Ion vs. VFB: Where Each Technology Wins

The choice isn't about "better"—it's about matching technology to operational requirements.

Lithium-Ion — Key Advantages

- **Lower upfront CAPEX**, mature supply chain
- **Higher energy density**, smaller footprint
- **85-90%+ round-trip efficiency**
- **Rapid deployment**, proven track record

Best fit: 1-4h duration, CAPEX-sensitive projects, frequency regulation

Vanadium Flow Battery — Key Advantages

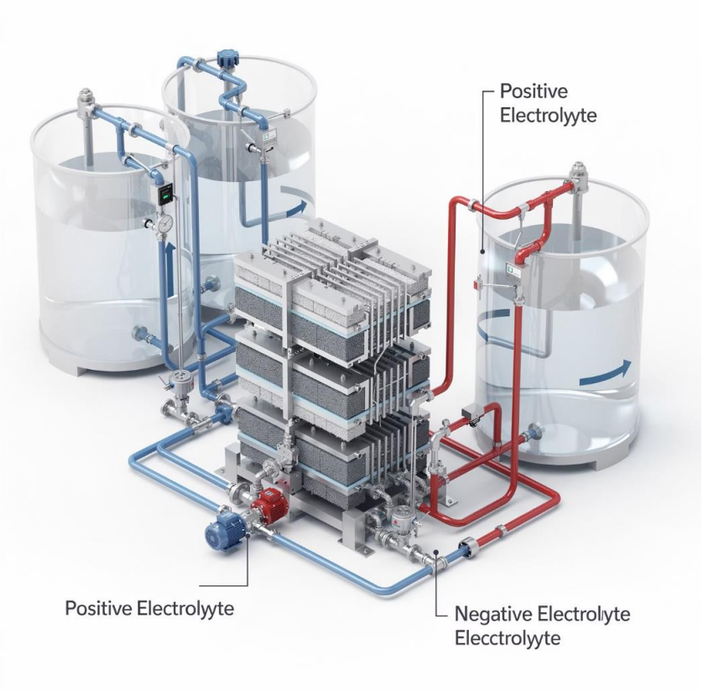
- **Intrinsic safety:** no thermal runaway risk
- **20,000+ deep cycles**, 25-year lifespan
- **100% depth of discharge** capability
- **Duration economics** improve at 4-10+ hours

Best fit: 4-10h+ duration, daily heavy cycling, safety-critical applications

Technology selection depends on duration, cycling intensity, safety requirements, and lifetime economics. The market increasingly recognizes that different grid services require different storage solutions.

VANADIUM FLOW

Why Vanadium Flow Battery Technology Exists



Fundamental Architecture

Vanadium Flow Batteries (VFB) use a fundamentally different approach than solid-state lithium-ion systems. The core innovation: **energy storage is decoupled from power delivery**.

The system consists of two main components:

- **Electrolyte tanks** containing vanadium solution—these determine energy capacity (MWh)
- **Power stacks** where the electrochemical reaction occurs—these determine power output (MW)

Scale Energy

Add more tanks or increase electrolyte volume to extend discharge duration (MWh)

Scale Power

Add more stacks to increase output capacity (MW) independently of energy storage

Grid Integration

Connect to transmission infrastructure with precise control over charge/discharge profiles

This architectural separation creates a decisive advantage for long-duration projects: you can economically optimize the MW/MWh ratio for specific applications. A 50 MW / 500 MWh system (10 hours) becomes feasible where lithium-ion economics would struggle. The electrolyte doesn't degrade with cycling, meaning the energy capacity remains stable over 20+ year lifetimes.

Section 5

Brazil Market Opportunities

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Primary Opportunities in Brazil: Where Capital and Pain Points Intersect

The Brazilian storage market isn't theoretical—specific applications with identifiable off-takers and economic drivers are emerging. The following four segments represent the highest-probability near-term opportunities for development and deployment.

LRCAP Peak Capacity Auction

Regulated product with contracted revenues and long-term price visibility. This is the "institutional entry point" providing bankable cash flows and established procurement processes. Expected to drive initial market development and set reference pricing.

Renewable Hybrid Projects in Northeast

ONS studies project significant curtailment risk in the 2026-2029 horizon as transmission capacity lags renewable additions. Storage collocated with wind and solar can mitigate curtailment losses while firming delivery profiles—directly improving project economics.

Market example: CGN + Goldwind deploying pilot BESS with wind farms to reduce curtailment exposure.

Transmission & Distribution Reinforcement

Storage as a non-wires alternative to traditional substation upgrades and transmission expansion. Already proven in Brazil with the Registro/SP installation—the country's first large-scale transmission-connected BESS.

This application offers unique advantages: avoid multi-year permitting for new lines, defer capital-intensive projects, and provide flexibility as load patterns evolve.

Commercial & Industrial Microgrids

Mission-critical operations requiring resiliency, power quality, and demand charge management. Key sectors: ports, mining operations, data centers, and continuous process manufacturing.

Storage becomes operational insurance—enabling continued operation during grid disturbances while optimizing energy costs through time-of-use arbitrage.

These four segments aren't mutually exclusive—leading developers will likely pursue portfolio strategies across multiple applications to diversify revenue streams and technology risk. The common thread: all represent situations where storage solves real operational problems with quantifiable economic value.

REGULATORY FRAMEWORK

LRCAP 2026: Contracting Reliability, Not Just Energy

Brazil's 2026 Capacity Reserve Auction represents a structural shift in the power sector.

What Is Contracted

AVAILABLE POWER (MW), not produced energy (MWh). The system pays for capacity to deliver power when needed, regardless of actual dispatch.

Regulatory Structure

MME
Auction guidelines, public policy

ANEEL
Regulates, tenders, oversees

ONS
Operates, dispatches assets

EPE
Studies, project registration

Key Requirements

- Minimum: 30 MW / 4-hour duration
- Recharge: within 6-hour window
- Mandatory ONS dispatch compliance
- Performance-based penalties

CRCAP Contract

- Duration: ~10 years
- Remuneration: R$/MW.year (availability-based)
- Strict performance obligations

LRCAP 2026

LRCAP 2026 Timeline: From Regulation to Operation

Indicative Timeline and Regulatory Milestones for Brazil's First Storage Auction

1. **November 7, 2025**
   MME Ordinance No. 878/2025: Guidelines published and Public Consultation No. 202/2025 opened
2. **November 10 - December 1, 2025**
   Public Consultation contribution period
3. **December 5, 2025**
   Joint Technical Note ANEEL-ONS-EPE published (metodology and technical criteria)
4. **January 2026**
   Technical Note on SIN Remaining Capacity published
5. **Q1 2026**
   Final Tender published by ANEEL
6. **April 2026**
   LRCAP 2026 – Storage auction
7. **2026-2028**
   Implementation and commissioning period for winning projects
8. **August 2028**
   Contractual supply start date (projected)
9. **2028-2038**
   Operation under CRCAP contract (~10 years)

Section 6

Exploring a Strategic Partnership: Lyon Capital and RKP

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RKP: Global Leader in Vanadium Flow Battery Platform

RKP has established itself as the dominant global player in vanadium flow battery technology, with operational scale, intellectual property depth, and reference projects that few competitors can match. This isn't emerging technology—it's proven infrastructure deployed at gigawatt-hour scale.

3+ GWh

Installed Capacity

Cumulative VFB deployments globally, demonstrating operational track record at utility scale

450+

Patents Held

Comprehensive IP portfolio covering VFB technology, electrolyte chemistry, and system integration

>80%

Market Share

Dominant position in VFB electrolyte supply globally—critical supply chain control

>20,000

Cycle Capability

Deep discharge cycles with minimal degradation, supporting 20-25 year project lifetimes

Flagship Reference Projects

Dalian, China

200 MW / 800 MWh installation (Phase I: 100 MW / 400 MWh)

World's largest single-location flow battery system, demonstrating grid-scale deployment capability and multi-hour discharge economics. Operational performance data validates technology reliability and dispatch flexibility.

Ushi, China

175 MW / 700 MWh with grid-forming capability

Large-scale grid-forming BESS demonstrating advanced grid support services beyond energy shifting. Critical reference for Brazil's LRCAP grid-forming requirements and transmission-connected applications.

These reference projects aren't pilots—they represent commercial-scale infrastructure providing essential grid services in competitive power markets. The operational data and performance validation from these installations directly de-risk technology selection for Brazilian applications, particularly for the 4-10+ hour duration segments where VFB economics are most compelling.

PARTNERSHIP STRATEGY

Exploring Strategic Partnership with RKP

Lyon Capital sees significant potential in vanadium flow battery technology for Brazil's emerging storage market—but the path forward requires deeper strategic alignment with RKP.

Critical Questions to Address:

RKP's Brazil Market Interest

- Strategic vision and priority level for Latin America
- Market conditions required for commitment
- Timeline for market entry

Technology Competitiveness

- VFB vs. Lithium-Ion: Economics for LRCAP 2026 and Other Market Opportunities
- Competitive CAPEX at Brazilian project scale

Business Model & Value Chain

The partnership structure should be defined based on RKP's strategic objectives and risk appetite. Potential models range from:

- Pure equipment supply with technology transfer
- Integrated EPC and commissioning participation
- Joint venture for long-term O&M and asset management
- Strategic co-investment in project SPVs
- Hybrid models combining multiple elements

What Lyon Capital Brings:

- Proven track record in energy infrastructure project development and implementation
- Deep expertise in Brazilian regulatory framework and market dynamics
- Established relationships with off-takers, utilities, and financing partners

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